

A Product-Form Counterexample Answers Q3 and Q5

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Status. The two questions below have negative answers, implied by a later explicit example of An–Lai [2]. This note records the exact identification; it does not claim a new result.

The source questions

Dutkay–Haussermann–Lai [1, Section 10, Q3 and Q5, PDF p. 38] ask, for an expansive integer matrix R and integer digit set B :

Q3. If the self-affine measure $\mu(R, B)$ is spectral, must there be L such that (R, B, L) is a Hadamard triple?

Q5. If $\mu(R, B)$ is spectral and has no overlap, must such an L exist?

The later example

An–Lai [2, Example 8.4, PDF p. 33] take

$$N = 24, \quad D = \{0, 1, 16, 17\}.$$

Their Theorem 2.6 proves that $\mu_{24,D}$ is a spectral measure. Example 8.4 then observes that D is not a spectral set in the finite cyclic group $\mathbb{Z}_{24}$ and concludes explicitly that there is no integer set L for which $(24, D, L)$ is an ordinary Hadamard triple.

There is one extra point needed for Q5. The elements of D are pairwise distinct modulo 24, so D is a simple digit set for 24. Theorem 1.6 of the source paper [1, PDF p. 5] states that every simple digit set for an expansive integer matrix satisfies the no-overlap condition. Hence the same example obeys the additional hypothesis in Q5.

Proposition. *Questions Q3 and Q5 of [1] both have negative answers.*

Proof. For $(R, B) = (24, D)$, spectrality and failure of every ordinary Hadamard triple are supplied by [2, Theorem 2.6 and Example 8.4]. Simplicity of D modulo 24 and [1, Theorem 1.6] give no overlap. Thus the example refutes Q3 and also Q5. $\square$

Scope. This does not answer Q4, which asks whether spectrality by itself forces no overlap, nor the broad Fourier-frame question Q2.

References

- [1] D. Dutkay, J. Haussermann, and C.-K. Lai, *Hadamard triples generate self-affine spectral measures*, Trans. Amer. Math. Soc. 371 (2019), 1439–1481; [arXiv:1607.08024](#).
- [2] L.-X. An and C.-K. Lai, *Product-form Hadamard triples and its spectral self-similar measures*, [arXiv:2209.05616](#).